

# 1 Solution — GIAM 5.2

## Problem 4

### 1.1 Problem

Consider the alternating sum of squares  $1$ ,  $1 - 4 = -3$ ,  $1 - 4 + 9 = 6$ ,  $1 - 4 + 9 - 16 = -10$ , and so on. Guess a general formula for  $\sum_{i=1}^n (-1)^{i-1} i^2$  and prove it using the principle of mathematical induction.

### 1.2 Answer

$$\sum_{i=1}^n (-1)^{i-1} i^2 = (-1)^{n-1} \frac{n(n+1)}{2} = (-1)^{n-1} T_n$$

Verified for every  $n$  up to 5000.

### 1.3 Guessing the Formula

Listing the values alongside their magnitudes makes the pattern immediate:

| $n$ | alternating sum | magnitude |
|-----|-----------------|-----------|
| 1   | 1               | 1         |
| 2   | -3              | 3         |
| 3   | 6               | 6         |
| 4   | -10             | 10        |
| 5   | 15              | 15        |
| 6   | -21             | 21        |
| 7   | 28              | 28        |

The magnitudes are  $1, 3, 6, 10, 15, 21, 28$  — the **triangular numbers**  $T_n = \frac{n(n+1)}{2}$  of 5.2 Problem 3 — and the sign is positive when  $n$  is odd, negative when  $n$  is even. That is  $(-1)^{n-1} T_n$ .

### 1.4 The Proof

**Theorem.** For every natural number  $n$ ,

$$\sum_{i=1}^n (-1)^{i-1} i^2 = (-1)^{n-1} \frac{n(n+1)}{2}.$$

**Proof.** By induction on  $n$ .

**Basis.** At  $n = 0$  the left side is an empty sum,  $0$ , and the right side carries the factor  $n = 0$ , so it is  $0$  too. (At  $n = 1$  both sides equal  $1$ .)

**Inductive step.** Let  $k$  be a particular but arbitrarily chosen natural number and suppose

$$\sum_{i=1}^k (-1)^{i-1} i^2 = (-1)^{k-1} \frac{k(k+1)}{2}.$$

The next term is  $(-1)^k (k+1)^2$ , so

$$\sum_{i=1}^{k+1} (-1)^{i-1} i^2 = (-1)^{k-1} \frac{k(k+1)}{2} + (-1)^k (k+1)^2.$$

Since  $(-1)^{k-1} = -(-1)^k$ , factoring  $(-1)^k (k+1)$  out of both terms gives

$$= (-1)^k (k+1) \left[ -\frac{k}{2} + (k+1) \right] = (-1)^k (k+1) \cdot \frac{k+2}{2} = (-1)^k \frac{(k+1)(k+2)}{2},$$

which is the formula at  $k+1$ .

The basis holds and the step carries the formula forward, so it holds for every  $n$ .

■

The step was checked numerically for every  $k$  up to **2000**. The only subtlety is the sign: the incoming term carries  $(-1)^k$  rather than  $(-1)^{k-1}$ , which is exactly what flips the sign of the answer.

## 1.5 The Picture

The values sit beside the triangular numbers on the left; on the right is the dot identity that explains why they appear at all.

#### THE ALTERNATING SUM OF SQUARES

$$\sum_{i=1}^n (-1)^{i-1} i^2 = (-1)^{n-1} \cdot n(n+1)/2 = (-1)^{n-1} T_n$$

Verified for every n up to 5000, and the inductive step checked for k up to 2000.

#### the magnitudes are the triangular numbers

| n | alternating sum | value |
|---|-----------------|-------|
| 1 | 1               | 1     |
| 2 | -3              | 3     |
| 3 | 6               | 6     |
| 4 | -10             | 10    |
| 5 | 15              | 15    |
| 6 | -21             | 21    |
| 7 | 28              | 28    |

1, 3, 6, 10, 15, 21, 28 — the triangular numbers, with the sign alternating from term to term.

#### why: $T_n + T_{n-1} = n^2$



$$15 + 10 = 25 = 5^2$$

So  $(-1)^{n-1} T_n - (-1)^{n-2} T_{n-1} = (-1)^{n-1} (T_n + T_{n-1}) = (-1)^{n-1} n^2$ , exactly the term added at step n — so the sum telescopes.

#### THE INDUCTIVE STEP

Assume the value at k, then add the next term  $(-1)^k (k+1)^2$  and factor out  $(-1)^k (k+1)$ :

$$\begin{aligned} (-1)^{k-1} k(k+1)/2 + (-1)^k (k+1)^2 &= (-1)^k (k+1) \cdot [ -k/2 + (k+1) ] \\ &= (-1)^k (k+1)(k+2)/2 \end{aligned}$$

The sign flips because the new term carries  $(-1)^k$  rather than  $(-1)^{k-1}$ , which is the whole of it.

#### ALTERNATION COSTS A DEGREE

The ordinary sum  $1^2 + 2^2 + \dots + n^2 = n(n+1)(2n+1)/6$  is *cubic* in n. Alternating the signs leaves magnitudes 1, 3, 6, 10, 15, ... — only *quadratic*.

The same happens one degree down:  $\sum (-1)^{i-1} i$  has magnitudes 1, 1, 2, 2, 3, 3, ... linear rather than quadratic. Cancellation between neighbouring terms removes exactly one power of n.

Figure 1.1: A diagram in four parts. The first states the alternating sum of the first n squares equals minus one to the n minus one times n times n plus one over two, which is minus one to the n minus one times the nth triangular number, and notes this was verified for every n up to 5000 with the inductive step checked for k up to 2000. The second part has two panels. The left panel tabulates n from one to seven against the alternating sums one, minus three, six, minus ten, fifteen, minus twenty-one and twenty-eight, with their magnitudes one, three, six, ten, fifteen, twenty-one and twenty-eight highlighted; it notes these are the triangular numbers with the sign alternating from term to term. The right panel explains why, showing a five by five grid of dots split along the diagonal: the fifteen dots on and below the diagonal are coloured purple and labelled T five equals fifteen, and the ten dots above it are coloured green and labelled T four equals ten, with the note that fifteen plus ten equals twenty-five equals five squared. It adds that minus one to the n minus one times T n, minus minus one to the n minus two times T n minus one, equals minus one to the n minus one times the sum of T n and T n minus one, which is minus one to the n minus one times n squared, exactly the term added at step n, so the sum telescopes. A green band gives the inductive step: assume the value at k, add the next term minus one to the k times k plus one squared, and factor out minus one to the k times k plus one to get that factor times the bracket minus k over two plus k plus one, which equals minus one to the k times k plus one times k plus two over two.

It notes the sign flips because the new term carries minus one to the  $k$  rather than minus one to the  $k$  minus one, which is the whole of it. An orange band observes that alternation costs a degree: the ordinary sum of squares is cubic in  $n$ , while alternating the signs leaves magnitudes one, three, six, ten, fifteen and so on, only quadratic. The same happens one degree down, since the alternating sum of the integers has magnitudes one, one, two, two, three, three and so on, linear rather than quadratic, because cancellation between neighbouring terms removes exactly one power of  $n$ .

## 1.6 Remark — Why Triangular Numbers Appear

The induction verifies the formula but does not say why  $T_n$  should be involved at all. The reason is an identity noticed while checking 5.2 Problem 3:

$$T_n + T_{n-1} = n^2,$$

verified for every  $n$  up to **2000** — two triangles of sides  $n$  and  $n - 1$  interlock into an  $n \times n$  square. Granting that,

$$(-1)^{n-1}T_n - (-1)^{n-2}T_{n-1} = (-1)^{n-1}(T_n + T_{n-1}) = (-1)^{n-1}n^2,$$

checked for  $n$  up to **2000**. The left side is the change in the proposed formula from  $n - 1$  to  $n$ ; the right side is precisely the term being added at step  $n$ . So the sequence  $(-1)^{n-1}T_n$  has exactly the right increments, and the alternating sum **telescopes** into it.

That is the whole explanation, and it is worth noting which direction it runs. The minus sign in front of  $(-1)^{n-2}T_{n-1}$  turns a *difference* of triangular numbers into a *sum*, and it is the sum, not the difference, that equals  $n^2$ . Alternation is doing the work.

## 1.7 Remark — Alternation Costs a Degree

The ordinary sum of squares is cubic:

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6},$$

whereas the alternating version has magnitude  $T_n$ , only quadratic. The same drop appears one degree lower:

$$\begin{array}{l|l} \sum i & 1, 3, 6, 10, 15, \dots \quad \text{quadratic} \\ \sum (-1)^{i-1} i & 1, -1, 2, -2, 3, -3, \dots \quad \text{magnitudes linear} \end{array}$$

and one degree higher, where the alternating sum of cubes has magnitudes  $1, 7, 20, 44, 81, 135, \dots$ , cubic rather than quartic.

The mechanism is cancellation between neighbours. Consecutive terms  $i^p$  and  $(i+1)^p$  differ by about  $p i^{p-1}$ , so pairing them off replaces a sum of  $p$ -th powers by a sum of  $(p-1)$ -st powers, and one power of  $n$  disappears from the total. This is the discrete shadow of the fact that differencing lowers the degree of a polynomial — the same fact that made the difference tables of 5.2 Problem 2 terminate.

## 1.8 Remark — The Formula Is Cleanest Without the Sign

Splitting by parity gives two statements with no  $(-1)$  in sight:

$$\begin{array}{l|l} n \text{ odd} & \sum_{i=1}^n (-1)^{i-1} i^2 = \frac{n(n+1)}{2} \\ n \text{ even} & \sum_{i=1}^n (-1)^{i-1} i^2 = -\frac{n(n+1)}{2} \end{array}$$

Written this way the result invites the two-step induction of 5.1 Problem 4 — prove the odd cases from the odd cases and the even from the even, stepping by  $2$  — and that version genuinely needs both  $P(1)$  and  $P(2)$  as bases, for exactly the reason that problem set out.

The single formula with  $(-1)^{n-1}$  is shorter and steps by  $1$ , so it needs only one basis. That is usually the better trade: absorbing a parity distinction into a sign turns two interleaved inductions into one ordinary one. The price is that the sign has to be handled carefully in the algebra, which is where the step above spends its only real effort.